
Toscanini 03 buyer decision card

invfixturefluidics_hydraulic0e8d4aa7cfb8 · run r541-ranked-pkg-battery-a · maturity
EARLY_TECHNICAL_EVALUATION · 3D class SYSTEM_3D

What is it?

A recorded invention candidate (inv:fixture:fluidics_hydraulic:0:e8d4aa7cfb8) for the problem: •

Problem Id: fixture:fluidics_hydraulic:0

- Device: CSF shunt system
- Failure: proximal catheter obstruction by tissue
- Failure Mode: OBSTRUCTION
- Constraint: maintain long-term lumen patency and flow — the proposed intervention: • Mechanism: porous microstructure resists fluid-path tissue ingrowth. The proposed intervention realizes this mechanism by placing porous titanium proximal catheter tip in the shunt lumen at the failure site identified in the problem statement, where the governing physical effect produces sustained...

What is actually established?

The architecture survived the recorded adversarial challenge, adjudication and evolution stages; status AUTOMATED_INVENTION_CANDIDATE. Evidence by class: direct support: 0; partial support: 0; background: 0; analogy: 1; contradictory: 0; irrelevant: 0. Model-derived engineering content is labeled MODEL_DERIVED; nothing claims physical validation (no PHYSICAL_OBSERVATION exists in this package).

What is not established?

FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003 — 13 recorded unknowns, each with a resolution action in UNKNOWN_ROADMAP.json. No numeric design decision is possible until the blockers are sourced.

What could kill it?

the falsification rule recorded in the decisive experiment contract

Why does it matter?

reliable operation of CSF shunt system

What is different?

- Collision Novelty Risk: SEARCHED_NO_DIRECT_TITLE_MATCH
- Prior Art Status: NO_MATCH_FOUND
- Multi Source Directions: (empty record)

What would we build?

- WP-01: test article first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed); measurement physical bench/clinical experiment — NOT simulation
- WP-02: test article constraint-compliance envelope article; measurement constraint-compliance characterization
- WP-03: test article article for VF-001; measurement bench shunt flow loop; measure flow decay over 30 days
- WP-04: test article article for VF-002; measurement run killer experiment:

falsification_test_from_candidate (EIG/cost 0.44573539382492045)

- WP-05: test article article for VF-003; measurement bench flow-loop characterization across operating range
- WP-06: test article article for VF-004; measurement pressure-drop mapping vs flow (identify regime transitions)
- WP-07: test article article for VF-005; measurement delivered-flow trend below baseline in a flow-loop or animal model; pressure-head rise at constant flow (measured quanti...
- WP-08: test article article for VF-006; measurement kink/collapse test across duty cycle; flow-stoppage signature under applied load (measured quantity: mechanical stress / ...
- WP-09: test article article for VF-007; measurement pressure-flow mapping across the full envelope showing regime-break curvature (measured quantity: pressure / pressure-dr...
- WP-10: test article article for VF-008; measurement reverse-leakage rate measurement after accelerated cycling (measured quantity: reverse-flow / leakage rate)

What must the buyer bring?

canonical problem context: MEDICAL_IN_VIVO

buyer type: medical device company (candidate buyer class for an in-vivo medical technology)

[APPLICABLE]

engineering capability: medical device design engineering with the domain's physics (see engineering specification governing model) [APPLICABLE]

manufacturing capability: medical-grade manufacturing (cleanroom assembly, medical extrusion/molding as the device requires) under a certified QMS [APPLICABLE]

manufacturing capability: sterilization-validation capability [APPLICABLE]

regulatory: medical-device regulatory pathway determination (FDA 510(k)/PMA or market jurisdiction equivalent) with biocompatibility evaluation (ISO 10993) and QMS (ISO 13485) [APPLICABLE]

market channels: clinical/hospital market channels [APPLICABLE]

What is the cheapest decisive next experiment?

physical bench/clinical experiment — NOT simulation

What should we do next?

Fund and run the recorded decisive experiment (physical bench/clinical experiment — NOT simulation) — the pre-registered rule either kills the architecture or graduates it to a sourced measurement.

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.